

Adaptive and Continual Learning for Human Activity Recognition

Chapter: Proposed System

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ESI Algeria, 2025/2026

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Chapter 1

Proposed System

1.1 Problem Formulation

Let $\mathcal{D} = \{(\mathbf{x}_i, y_i, s_i)\}_{i=1}^N$ be a dataset of inertial signals, where $\mathbf{x}_i \in \mathbb{R}^{T \times C}$ is a time-series window of T timesteps and C sensor channels, $y_i \in \mathcal{Y}$ is the activity label, and s_i is the subject identifier.

We address two complementary objectives:

1. **Continual Activity Recognition.** Given a sequence of tasks $\{\mathcal{T}_1, \mathcal{T}_2, \dots, \mathcal{T}_K\}$ arriving over time—each representing a new user, a new activity class, or a new context—train a model f_θ that performs well on *all* tasks seen so far without retraining from scratch and without forgetting previously acquired knowledge (*catastrophic forgetting*).
2. **Activity Anticipation.** Given a partial observation $\mathbf{x}^{(p)} \in \mathbb{R}^{pT \times C}$ ($p \in (0, 1)$) of an ongoing activity, predict the label $\hat{y}^{(+1)}$ of the *next* activity before it begins.

1.2 Data Preprocessing Pipeline

Raw inertial signals collected from heterogeneous datasets differ in sampling frequency, unit of measurement, and presence of gravitational acceleration. We apply a three-step homogenisation procedure inspired by [1].

1.2.1 Signal Homogenisation

Resampling. All signals are resampled to a common frequency $f_s = 50$ Hz using Fourier-based resampling. The number of output samples is:

$$N_{\text{new}} = \left\lfloor N_{\text{orig}} \times \frac{f_s}{f_{\text{orig}}} \right\rfloor \quad (1.1)$$

Unit Conversion. Accelerometer channels expressed in gravitational units (g) are converted to SI units (m/s^2):

$$\mathbf{a}_{[\text{m/s}^2]} = \mathbf{a}_{[g]} \times g_0, \quad g_0 = 9.80665 \text{ m/s}^2 \quad (1.2)$$

Gravity Removal. Gravity is estimated via a 5th-order Butterworth low-pass filter with cutoff $f_c = 0.5$ Hz and removed by subtraction:

$$\tilde{\mathbf{a}}(t) = \mathbf{a}(t) - \mathcal{F}_{\text{LP}}(\mathbf{a})(t) \quad (1.3)$$

where $\mathcal{F}_{\text{LP}}$ denotes the low-pass filter operator. This step is applied uniformly regardless of whether gravity is declared present in the dataset metadata [1].

1.2.2 Label Homogenisation

Each dataset uses its own activity vocabulary. We define a unified label space $\mathcal{Y}_u$ of 12 activity classes (Table 1.1) and map each dataset-specific label to the closest semantic equivalent using a combination of syntactic string similarity (Levenshtein distance) and signal-level similarity (Euclidean distance between hand-crafted feature vectors).

Table 1.1: Unified activity label space $\mathcal{Y}_u$.

ID	Activity	Datasets	Windows
1	Walking	HAPT, WISDM, PAMAP2	17,349
2	Walking Upstairs	HAPT, PAMAP2	6,374
3	Walking Downstairs	HAPT, PAMAP2	5,477
4	Sitting / Standing	HAPT, WISDM, PAMAP2	9,593
5	Jogging / Running	WISDM, PAMAP2	12,048
6	Lying	HAPT, PAMAP2	3,112
7	Cycling	PAMAP2	1,095
8	Nordic Walking	PAMAP2	1,255
9	Stand $\rightarrow$ Sit	HAPT	138
10	Sit $\rightarrow$ Stand	HAPT	109
11	Sit $\rightarrow$ Lie	HAPT	163
12	Lie $\rightarrow$ Sit	HAPT	148

1.2.3 Sliding Window Segmentation

Each preprocessed signal is divided into overlapping windows of fixed length $T = f_s \times W_{\text{sec}} = 50 \times 3 = 150$ samples with a stride of $S = T \times (1 - r_{\text{ov}}) = 75$ samples ($r_{\text{ov}} = 0.5$). The label of each window is determined by majority vote over its sample-level labels:

$$y_w = \arg \max_{c \in \mathcal{Y}_u} \sum_{t=t_0}^{t_0+T-1} \mathbf{1}[y(t) = c] \quad (1.4)$$

This yields $N = 56,861$ labelled windows across 44 subjects from three publicly available datasets (HAPT, WISDM, PAMAP2).

1.3 Model Architecture

The proposed system consists of a shared *backbone* encoder and two task-specific heads, as illustrated in Figure 1.1.

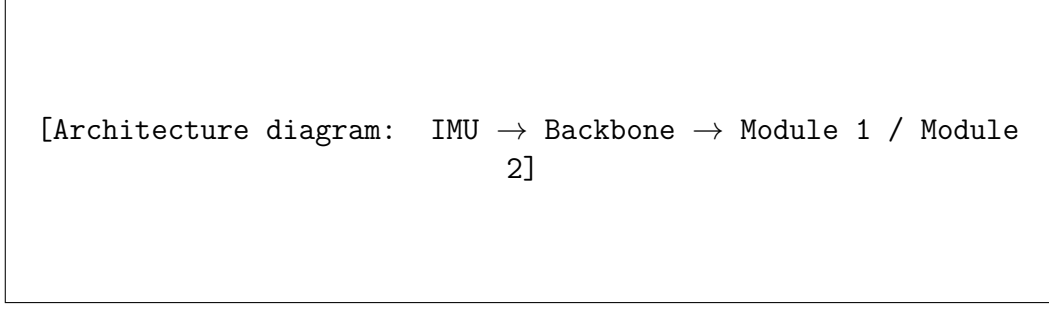


Figure 1.1: Overview of the proposed two-headed continual HAR architecture.

1.3.1 IMU Transformer Encoder (Backbone)

Inspired by [4], we adopt a Transformer encoder as the shared feature extractor. The backbone maps each window $\mathbf{x} \in \mathbb{R}^{T \times C}$ to a fixed-length embedding vector $\mathbf{e} \in \mathbb{R}^d$.

Input Projection. A linear layer followed by Layer Normalisation projects the raw C -channel input to the model dimension d :

$$\mathbf{H}^{(0)} = \text{LayerNorm}(\mathbf{x} \mathbf{W}_{\text{in}} + \mathbf{b}_{\text{in}}), \quad \mathbf{W}_{\text{in}} \in \mathbb{R}^{C \times d} \quad (1.5)$$

CLS Token. A learnable class token $\mathbf{v}_{\text{cls}} \in \mathbb{R}^d$ is prepended to the projected sequence:

$$\tilde{\mathbf{H}}^{(0)} = [\mathbf{v}_{\text{cls}}; \mathbf{H}^{(0)}] \in \mathbb{R}^{(T+1) \times d} \quad (1.6)$$

Positional Encoding. A combination of fixed sinusoidal and learnable positional encodings is added:

$$\mathbf{H} = \tilde{\mathbf{H}}^{(0)} + \lambda \cdot \mathbf{P}, \quad P_{t,2k} = \sin\left(\frac{t}{10000^{2k/d}}\right), \quad P_{t,2k+1} = \cos\left(\frac{t}{10000^{2k/d}}\right) \quad (1.7)$$

where $\lambda \in \mathbb{R}$ is a learnable scaling parameter initialised to 1.

Transformer Encoder Blocks. The sequence is processed by $L = 4$ identical encoder blocks. Each block applies pre-normalisation multi-head self-attention followed by a position-wise feed-forward network (pre-norm residual):

$$\mathbf{Z}^{(\ell)} = \mathbf{H}^{(\ell-1)} + \text{MHA}(\text{LN}(\mathbf{H}^{(\ell-1)})) \quad (1.8)$$

$$\mathbf{H}^{(\ell)} = \mathbf{Z}^{(\ell)} + \text{FFN}(\text{LN}(\mathbf{Z}^{(\ell)})) \quad (1.9)$$

Multi-head attention with $H = 4$ heads computes:

$$\begin{aligned} \text{MHA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) &= \text{Concat}(\text{head}_1, \dots, \text{head}_H) \mathbf{W}^O, \\ \text{head}_h &= \text{softmax}\left(\frac{\mathbf{Q} \mathbf{W}_h^Q (\mathbf{K} \mathbf{W}_h^K)^\top}{\sqrt{d/H}}\right) \mathbf{V} \mathbf{W}_h^V \end{aligned} \quad (1.10)$$

The feed-forward sub-layer uses GELU activation and inner dimension $d_{\text{ff}} = 2d = 256$:

$$\text{FFN}(\mathbf{z}) = \text{GELU}(\mathbf{z} \mathbf{W}_1 + \mathbf{b}_1) \mathbf{W}_2 + \mathbf{b}_2 \quad (1.11)$$

Output. The final embedding is the CLS token of the last block after Layer Normalisation:

$$\mathbf{e} = \text{LN}(\mathbf{H}^{(L)})_{[0]} \in \mathbb{R}^d \quad (1.12)$$

The backbone contains 531,457 trainable parameters with $d = 128$.

1.3.2 Module 1 — Continual HAR Head

Prototype Memory

We adopt a *prototype-based* nearest-mean classifier inspired by LAPNet-HAR [2]. Each activity class $c \in \mathcal{Y}$ is represented by a prototype vector $\boldsymbol{\mu}_c \in \mathbb{R}^d$, defined as the running mean of all embeddings observed for that class:

$$\boldsymbol{\mu}_c^{(n+m)} = \frac{n \boldsymbol{\mu}_c^{(n)} + m \bar{\mathbf{e}}_c^{\text{new}}}{n + m} \quad (1.13)$$

where n is the previous count, m the new batch count, and $\bar{\mathbf{e}}_c^{\text{new}}$ the mean embedding of the incoming batch. This update is performed online without storing past embeddings.

Nearest-Mean Classification. At inference, a window $\mathbf{x}$ is classified by finding the closest prototype in Euclidean space:

$$\hat{y} = \arg \min_{c \in \mathcal{K}} \|\mathbf{e} - \boldsymbol{\mu}_c\|_2^2 \quad (1.14)$$

where $\mathcal{K}$ is the set of all classes seen so far. This *fixed-capacity* head accommodates new classes without modifying the network architecture.

Experience Replay Buffer

To mitigate catastrophic forgetting, we maintain a replay buffer $\mathcal{M}$ of capacity $|\mathcal{M}| = 2,000$ windows using class-balanced reservoir sampling. Letting $q = \lfloor |\mathcal{M}| / |\mathcal{K}| \rfloor$ denote the per-class quota, each class retains at most q samples. When a new sample $(\mathbf{x}, y)$ arrives for class c , it replaces an existing buffer entry with probability:

$$p_{\text{replace}} = \frac{|\mathcal{M}| / |\mathcal{K}|}{n_c + 1} \quad (1.15)$$

where n_c is the total number of samples seen for class c . During continual training, a replay batch $\mathcal{B}_r \sim \mathcal{M}$ is mixed with the current task batch $\mathcal{B}_t$ at every update step.

Supervised Contrastive Loss

To enforce inter-class separation in the embedding space as new classes arrive, we apply the supervised contrastive loss of [3]. Given a batch of B ℓ_2 -normalised embeddings $\hat{\mathbf{e}}_i = \mathbf{e}_i / \|\mathbf{e}_i\|_2$:

$$\mathcal{L}_{\text{contrast}} = -\frac{1}{B} \sum_{i=1}^B \frac{1}{|\mathcal{P}(i)|} \sum_{j \in \mathcal{P}(i)} \log \frac{\exp(\hat{\mathbf{e}}_i^\top \hat{\mathbf{e}}_j / \tau)}{\sum_{k \neq i} \exp(\hat{\mathbf{e}}_i^\top \hat{\mathbf{e}}_k / \tau)} \quad (1.16)$$

where $\mathcal{P}(i) = \{j \neq i : y_j = y_i\}$ is the set of positive pairs for anchor i and $\tau = 0.07$ is the temperature.

Continual Training Objective

The total loss at each continual update step combines three terms:

$$\mathcal{L}_{\text{total}} = \underbrace{\mathcal{L}_{\text{CE}}(\mathcal{B}_t)}_{\text{current task}} + \underbrace{\mathcal{L}_{\text{CE}}(\mathcal{B}_r)}_{\text{replay}} + \lambda_c \underbrace{\mathcal{L}_{\text{contrast}}(\mathcal{B}_t)}_{\text{separation}} \quad (1.17)$$

where $\mathcal{L}_{\text{CE}}$ is cross-entropy and $\lambda_c = 0.1$ controls contrastive regularisation strength.

After each task, prototype vectors are *re-adapted* to correct for the shift in the embedding space caused by backbone weight updates:

$$\boldsymbol{\mu}_c \leftarrow \frac{1}{|\mathcal{M}_c|} \sum_{\mathbf{x} \in \mathcal{M}_c} f_{\theta}(\mathbf{x}), \quad \forall c \in \mathcal{K} \quad (1.18)$$

This prevents *prototype obsolescence* [2].

1.3.3 Module 2 — Activity Anticipation Head

The anticipation head predicts the label of the *next* activity from a partial observation of the current one. Given a sequence of S partial windows $(\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(S)})$, each of length pT ($0 < p < 1$), the backbone produces a sequence of embeddings:

$$\mathbf{e}^{(k)} = f_{\theta}(\mathbf{x}^{(k)}) \in \mathbb{R}^d, \quad k = 1, \dots, S \quad (1.19)$$

These are processed by a 2-layer LSTM with hidden dimension $d_h = 128$:

$$\mathbf{h}^{(k)}, \mathbf{c}^{(k)} = \text{LSTM}(\mathbf{e}^{(k)}, \mathbf{h}^{(k-1)}, \mathbf{c}^{(k-1)}) \quad (1.20)$$

The final hidden state produces next-activity logits:

$$\hat{\mathbf{y}}^{(+1)} = \text{softmax}(\text{LN}(\mathbf{h}^{(S)}) \mathbf{W}_{\text{out}}) \in \mathbb{R}^{|\mathcal{Y}|} \quad (1.21)$$

The head is trained at three observation ratios $p \in \{0.25, 0.50, 0.75\}$ with cross-entropy loss, encouraging robustness to inputs of varying completeness.

1.4 Training Procedure

Training proceeds in two stages.

1.4.1 Stage 1 — Supervised Pre-training

The backbone and a linear softmax head are trained jointly on the full merged dataset using cross-entropy loss, AdamW [6] ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay = 10^{-4}), and cosine annealing:

$$\eta_t = \eta_{\min} + \frac{1}{2} (\eta_0 - \eta_{\min}) \left(1 + \cos \frac{\pi t}{T_{\max}} \right) \quad (1.22)$$

with $\eta_0 = 10^{-3}$, $\eta_{\min} = 10^{-5}$, $T_{\max} = 30$ epochs, batch size 128. Gradient norms are clipped at 1.0. After pre-training, prototype vectors are initialised from the training-set embeddings via Eq. (1.13).

1.4.2 Stage 2 — Continual Learning

Tasks arrive sequentially. For each task $\mathcal{T}_i$, the model is updated for $E = 10$ epochs using the composite loss in Eq. (1.17) with a reduced learning rate $\eta = 5 \times 10^{-5}$. After each task, prototypes are re-adapted via Eq. (1.18). Two scenarios are evaluated:

- **User-incremental:** each task $\mathcal{T}_i$ introduces one new subject s_i . The model adapts to individual movement styles while retaining performance across all prior subjects.
- **Class-incremental:** each task $\mathcal{T}_i$ introduces two new activity classes. The model extends its vocabulary without retraining from scratch.

1.5 Evaluation Metrics

Let $R_{i,j}$ denote the macro-F1 score on the test set of task $\mathcal{T}_j$ after training on task $\mathcal{T}_i$ ($i \geq j$). The matrix $\mathbf{R} \in \mathbb{R}^{K \times K}$ captures the complete performance profile across all tasks and training stages.

Macro F1 Score. Primary recognition metric, robust to class imbalance:

$$F1_{\text{macro}} = \frac{1}{|\mathcal{Y}|} \sum_{c \in \mathcal{Y}} \frac{2 \text{TP}_c}{2 \text{TP}_c + \text{FP}_c + \text{FN}_c} \quad (1.23)$$

Backward Transfer (BWT). Measures degradation of old tasks after new ones are learned [5]. Negative values indicate forgetting:

$$\text{BWT} = \frac{1}{K-1} \sum_{j=1}^{K-1} (R_{K,j} - R_{j,j}) \quad (1.24)$$

Forgetting. Average maximum accuracy drop per old task:

$$\mathcal{F} = \frac{1}{K-1} \sum_{j=1}^{K-1} \left(\max_{i \leq K} R_{i,j} - R_{K,j} \right) \quad (1.25)$$

Forward Transfer (FWT). Measures whether learning earlier tasks accelerates new ones:

$$\text{FWT} = \frac{1}{K-1} \sum_{i=2}^K (R_{i-1,i} - b_i) \quad (1.26)$$

where b_i is the random-chance baseline accuracy for task $\mathcal{T}_i$.

Anticipation Accuracy. Reported at each observation ratio $p \in \{0.25, 0.50, 0.75\}$ to characterise the trade-off between prediction lead time and accuracy.

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